

Trust, Transparency, and Control: LLMs in High-Stakes Financial Decisions

Hui Chen

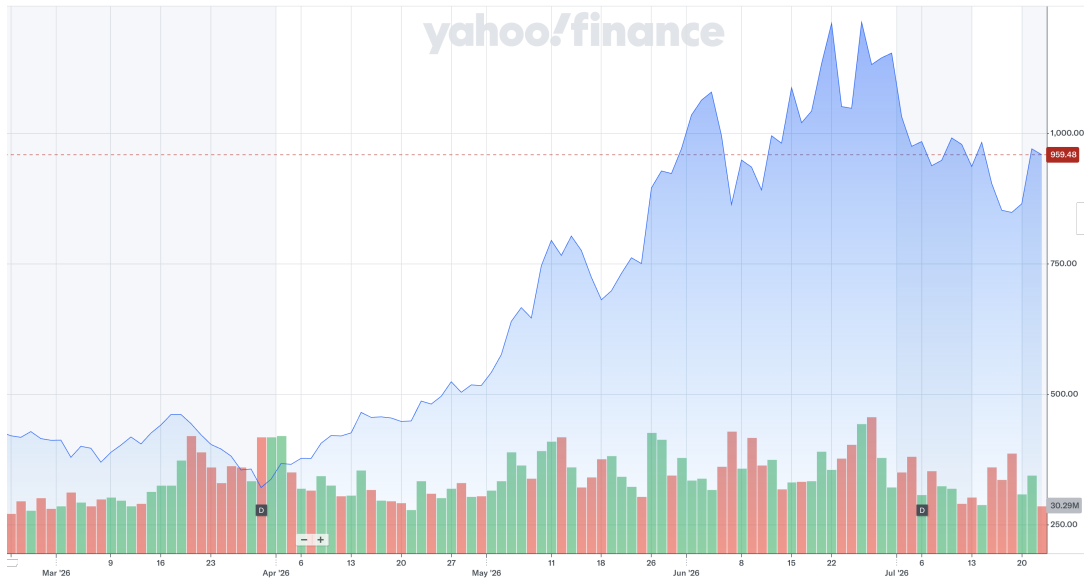
MIT Sloan

AI for Financial Services: Tools, Opportunities, and Challenges

Outline

- 1 Why Not the Black Box?
- 2 When To Trust the LLM?
- 3 Opening the Black Box
- 4 Human In the Loop
- 5 Closing

yahoo!finance



How much to trust the black box?

Prompt

Do you think MU will go up or down this coming week? Answer with up, down, or uncertain.

LLM: Up.

How much to trust the black box?

Prompt

Based on the information below, what do you think the headline CPI inflation will be in Q3 2026?

{news}

LLM: Q3 2026 headline CPI will be 3.3% year over year.

How much to trust the black box?

Prompt I

Give me a random number between 0 and 9

Prompt II

Give me a random number between 9 and 0

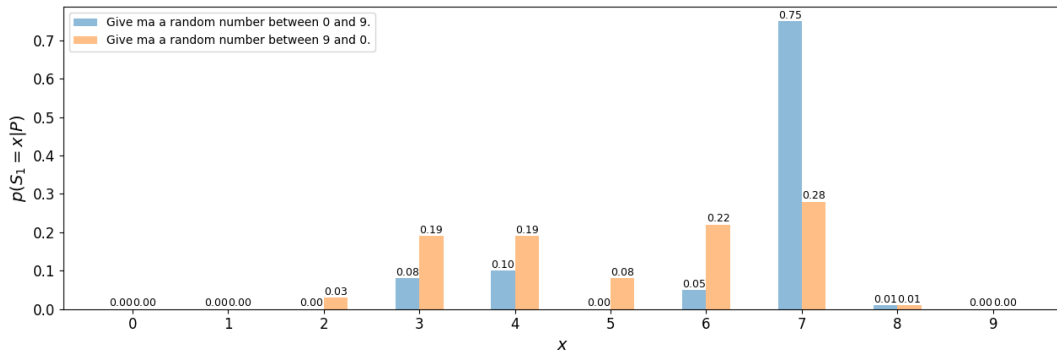
How much to trust the black box?

Prompt I

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Prompt II

Give me a random number between 9 and 0



What does this image remind you of?



Investors made a lot of profit in AAPL.

Do you want to invest in a stock or a bond? Only reply with 'stock' or 'bond'.
Do not reply with other answers. Your choice is:



Stock

What does this image remind you of?



Kobe Bryant lost championship to the Celtics.

Do you want to invest in a stock or a bond? Only reply with 'stock' or 'bond'.
Do not reply with other answers. Your choice is:



Bond

- Emotional content, even if irrelevant for the current decision domain, biases the results.

How much to trust the black box?



AI can produce a perfect-looking proof that is completely wrong, and most people have no way to catch it.

Terence Tao
Mathematician



There is an aspect of this which we call ... a black box. You know, you don't fully understand.

Sundar Pichai
CEO of Alphabet



Algorithms must not be a black box and there must be clear rules if something goes wrong.

Ursula von der Leyen
President, European Commission

Chatbots $\Rightarrow$ Agentic Systems $\Rightarrow$ Human-AI Collaboration

- LLMs are getting more powerful. But they are still deficient for high-stake decision-making.

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- When can AI be trusted?
 - ↪ Use the model's internal confidence to filter, escalate, and size decisions.

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 - $\hookrightarrow$ Use the model's internal confidence to filter, escalate, and size decisions.
- What is my AI's reasoning process?
 - $\hookrightarrow$ Use interpretability and steering to understand what drives outputs and control its behavior.

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- Will AI replace my employees?
 - $\hookrightarrow$ Finance requires domain knowledge and accountability: human-AI collaboration

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Turn AI confidence into a control

Paper

Out of the Black Box: Uncertainty Quantification for LLMs via Conditional Probabilities

- Main question: **When can we trust an LLM's answer?**
- Basic idea: Every LLM answer comes from a probability distribution over plausible tokens.
- Instead of looking only at the final text, pay attention to how concentrated or diffuse that internal distribution is.
- This gives us a native confidence signal that can sit behind routing, approval, and risk limits.

A primer on LLM

- Autoregressive LLM = a neural language model (transformer) + a decoding strategy
- The transformers are probabilistic models, providing the conditional probability distribution for the next token/word given the prompt tokens:

$$p(s | P) \quad s \in V$$

- Example: *Investing is ...*

$$p(\text{science} | \text{investment, is}) = 0.4$$

$$p(\text{art} | \text{investment, is}) = 0.2$$

$$p(\text{gambling} | \text{investment, is}) = 0.1$$

...

A primer on LLM

- The decoding strategy is an algorithm that selects the next token from the conditional distribution.

↪ One possible strategy is to select the token with the highest conditional probability (greedy decoding):

$$s_1 = \arg \max_{s \in V} p(s | P).$$

- ↪ Alternatively, we can sample the next token from the conditional distribution (stochastic decoding).
- ↪ Degree of randomness can be controlled by temperature.

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↪ Degree of randomness can be controlled by temperature.

- Once drawn, $\tilde{s}_1$ is appended to the end of the original prompt and affects the conditional distribution of the next token.

$$\textcolor{red}{s}_1 \leftarrow p(s | P)$$

$$\textcolor{blue}{s}_2 \leftarrow p(s | \textcolor{red}{s}_1, P)$$

$$\textcolor{violet}{s}_3 \leftarrow p(s | \textcolor{blue}{s}_2, \textcolor{red}{s}_1, P)$$

...

Uncertainty quantification

Prompt

Based on the news below, do you think MU's stock reaction will be positive or negative?

Output exactly one character: P or N, and nothing else.

{Micron just did in three days what once took 37 years}

{Micron has fully sold out of its 2026 inventory, and customers are signing multiyear prepayment agreements ...}

Uncertainty quantification

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{Micron has fully sold out of its 2026 inventory, and customers are signing
multiyear prepayment agreements ...}

Should we react the same way in the following two situations?

- 1 P has 90% probability and N has 10% probability.
- 2 P has 51% probability and N has 49% probability.

Uncertainty quantification

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Self-declared confidence

Prompt

Based on the news below, do you think the stock price reaction will be positive or negative?
Output exactly one character: P or N, and nothing else.

Then write how confident you are in your choice by providing a number between 1 (very confident) and 0 (not confident).

{headline}

{body}

Self-declared confidence

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{headline}

{body}

Can AI be trusted to self-assess its confidence?

How we validate this measure

Test 1: Does confidence predict accuracy?

- 100,000 Reuters firm-news items
- Intraday timestamp, all after the GPT-4o training cutoff
- Classify short-term return impact as positive or negative
- Ground truth: realized same-day open-to-close return

How we validate this measure

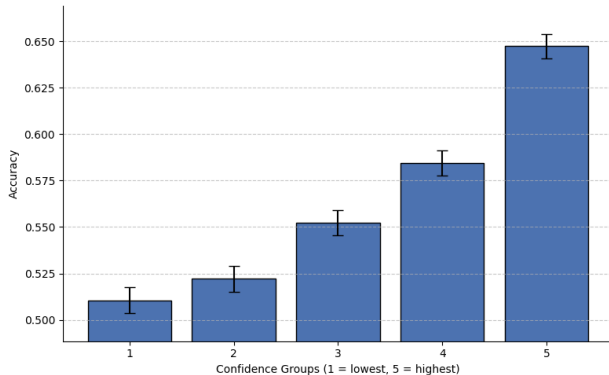
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Test 2: Does confidence improve return?

- Use overnight news to build a long-short strategy
- Split signals into **high** and **low confidence**
- Compare both against the unconditional baseline
- Does confidence enhance investment performance?

What our research finds



Higher confidence $\Rightarrow$ more accurate

- Top-confidence predictions reached about **64.7%** accuracy.
- The lowest-confidence bucket was about **51%**.

What our research finds

	Inner Confidence		Declared Confidence		Baseline
	High Conf	Low Conf	High Conf	Low Conf	
Mean (%)	49.62	-9.80	40.83	-3.55	39.56
Std (%)	9.63	25.19	11.01	32.83	9.34
Sharpe ratio	5.15	-0.39	3.71	-0.11	4.24
Avg # stocks	481.50	111.82	452.42	130.82	525.71

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Trading value

- High-confidence signals $\Rightarrow$ significantly higher Sharpe ratio. Low-confidence signals are essentially noise.
- Model-declared confidence does not add value.

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Trading value

- High-confidence signals $\Rightarrow$ significantly higher Sharpe ratio. Low-confidence signals are essentially noise.
- Model-declared confidence does not add value.
- Inner confidence is not yet properly calibrated.

Practical use cases

- Investment research
- News and filing processing
- Credit and underwriting review
- Internal knowledge assistants
- Human-in-the-loop agentic workflows

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A practical rule

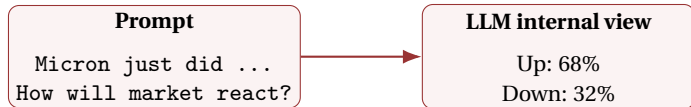
Know when (not) to trust the LLM.

A simple mental model

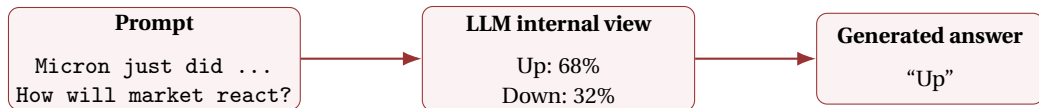
Prompt

Micron just did ...
How will market react?

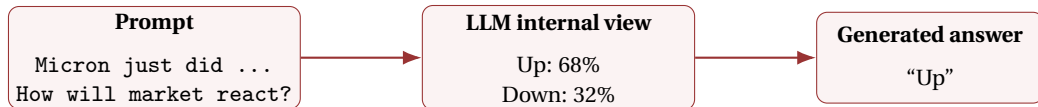
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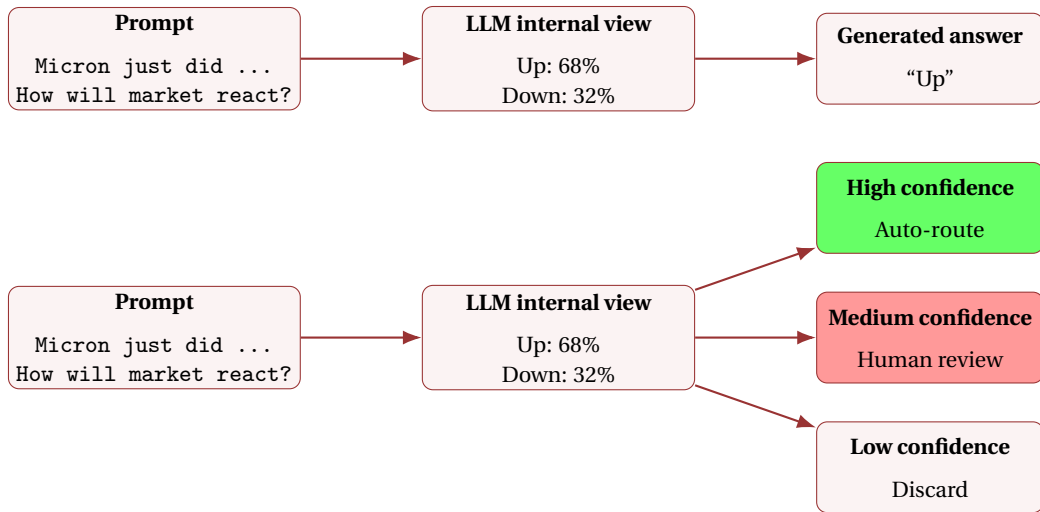
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Interpretability and Steering

Paper

A Financial Brain Scan of the LLM

- Main question: **What is the model's reasoning process when it makes a financial judgment?**
- Open a window into the model's internal reasoning and map it to interpretable financial concepts.

Needing Interpretability

Prompt

Based on the information below, what do you think the headline CPI inflation will be in Q3 2026?

{news}

LLM: Q3 2026 headline CPI will be 3.3% year over year.

Needing Control

Prompt

Recommend a portfolio allocation for my client, but be **20% more risk averse** than you have been previously.

Why interpretation is necessary in high-stakes finance

Performance alone is not enough

- A strong backtest does not tell us what the model is loading on
- It may be using future information or over-fitting the data
- Serious capital and high-impact decisions need more than a score

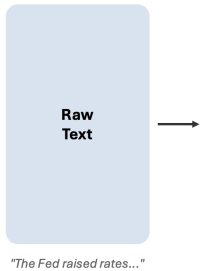
Prompting is not governance

- “Be cautious” is not a reliable control
- “Be optimistic” is not a measurable parameter
- In regulated or audited settings, we need something more systematic

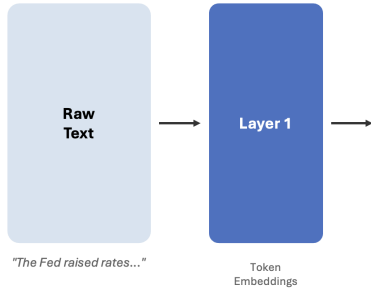
The core requirement

If an AI system is applied for high-stake decision-making, we need to know its reasoning process.

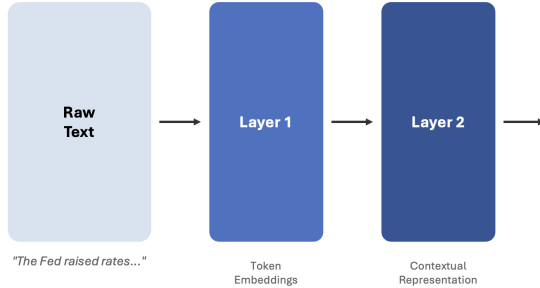
Looking inside the LLM



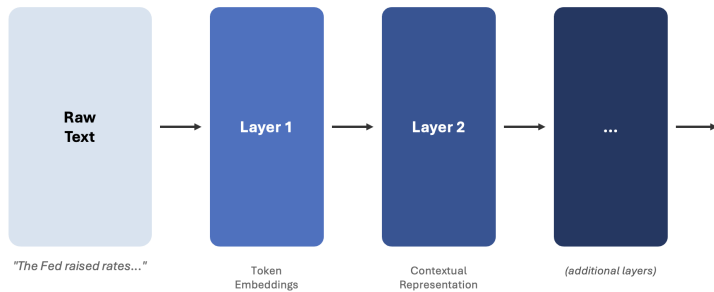
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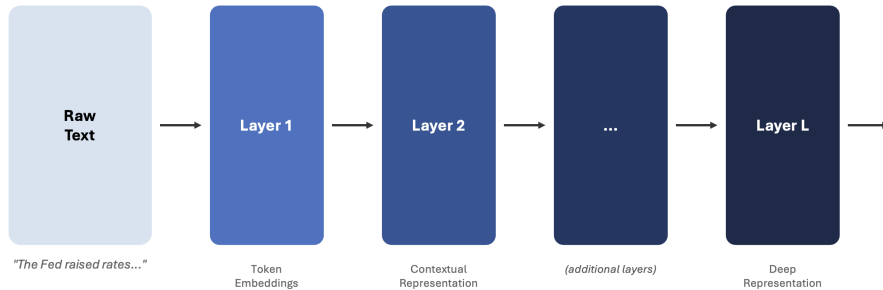
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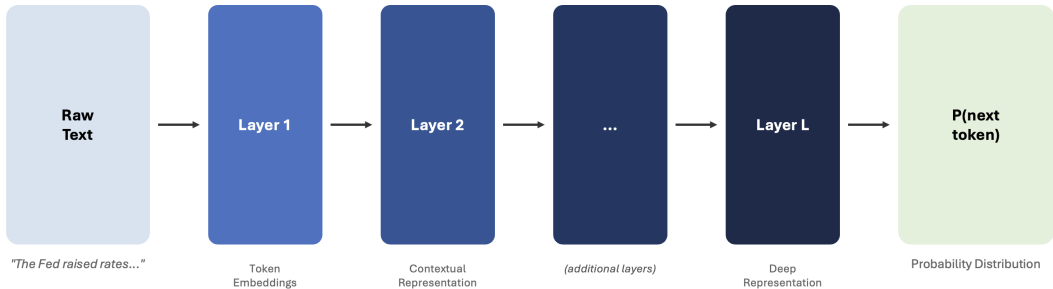
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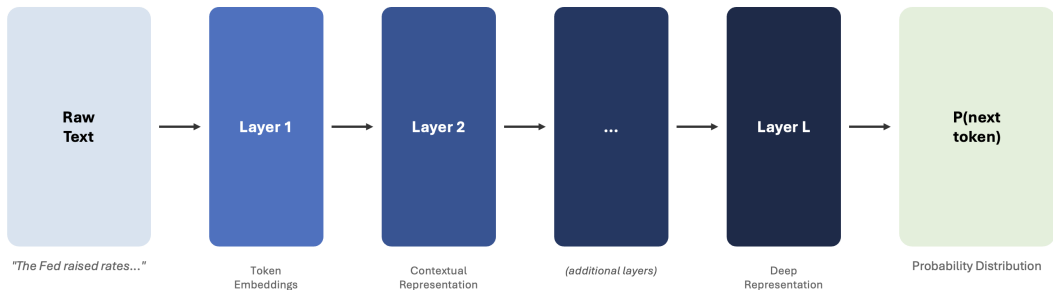
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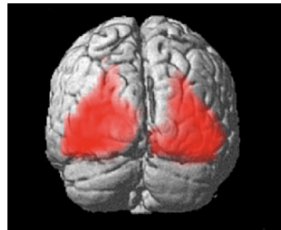
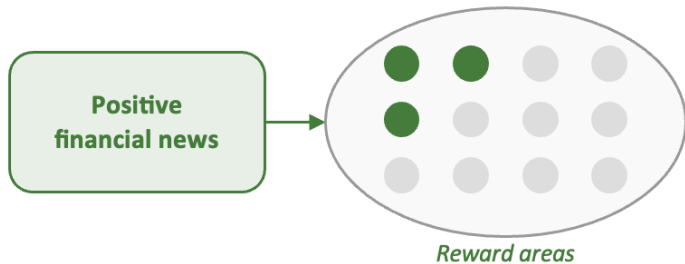


Looking inside the LLM

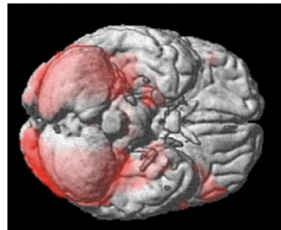
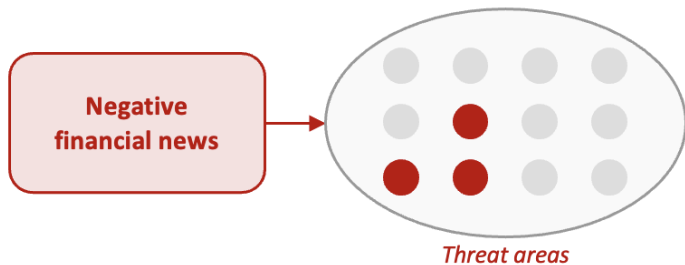


LLM builds progressively more useful representations of the context to predict the next token.

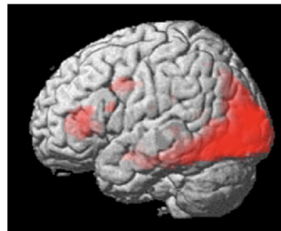
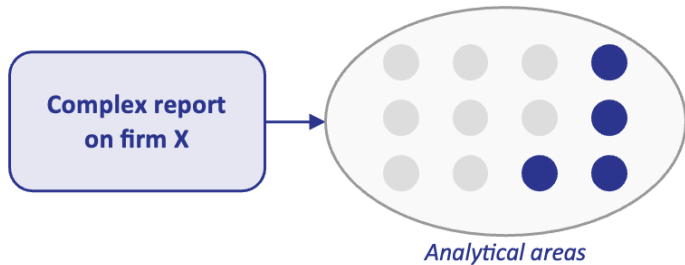
Can we perform a brain scan of the LLM?



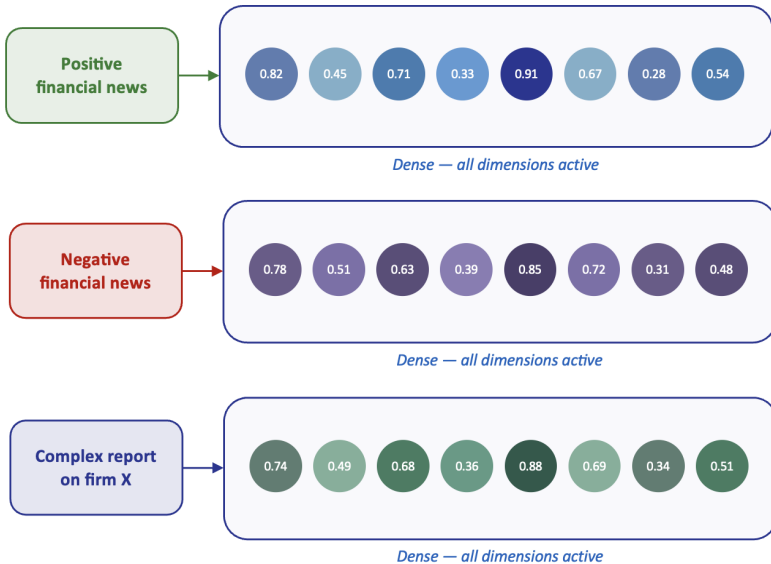
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LLM Embeddings



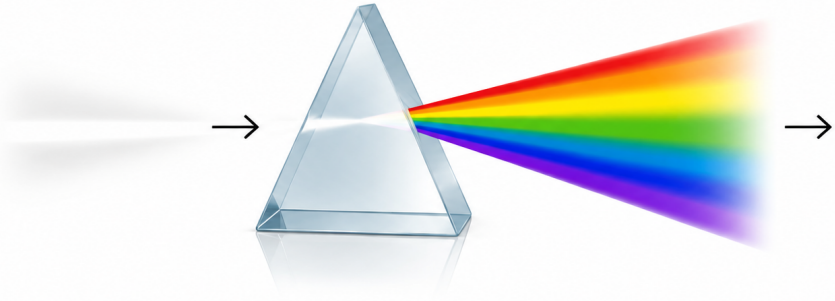
Dispersive Prism

1. Incoming
White Light

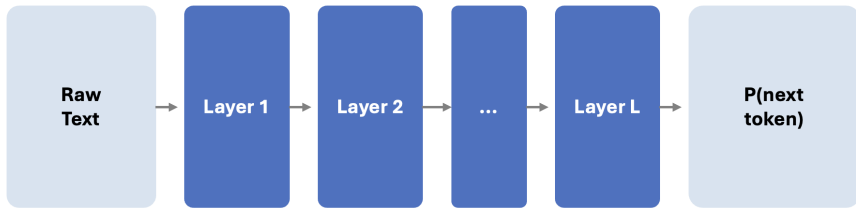
2. Dispersive
Prism

3. Dispersed
Spectrum

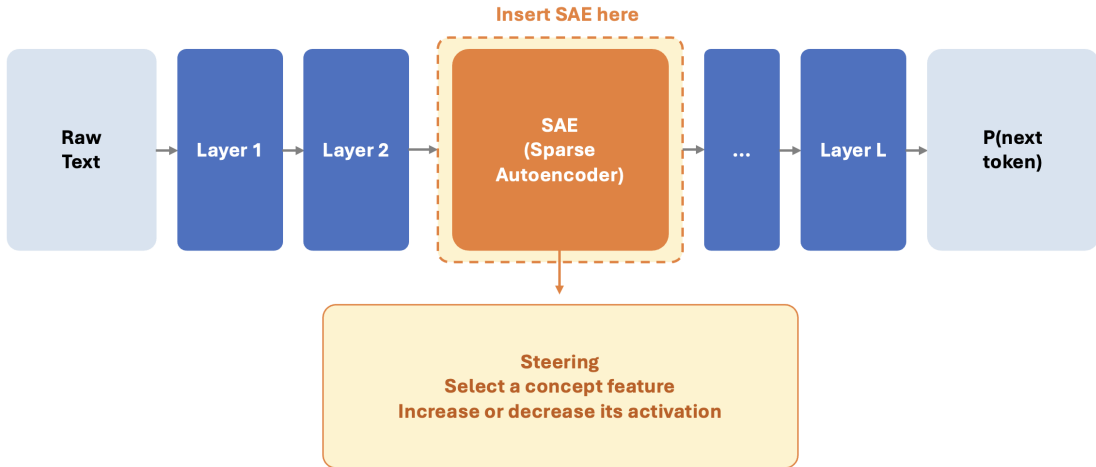
4. Interpretation:
Wavelengths Revealed

- 
- The diagram illustrates the process of light dispersion. A beam of white light enters a triangular glass prism from the left. As the light exits the right side of the prism, it is dispersed into a fan-shaped spectrum of colors. The colors transition from red at the top to violet at the bottom. To the right of the spectrum, a legend lists the colors with corresponding wavy lines: Red (long λ), Orange, Yellow, Green, Blue, Indigo, and Violet (short λ).
- Red (long λ)
 - Orange
 - Yellow
 - Green
 - Blue
 - Indigo
 - Violet (short λ)

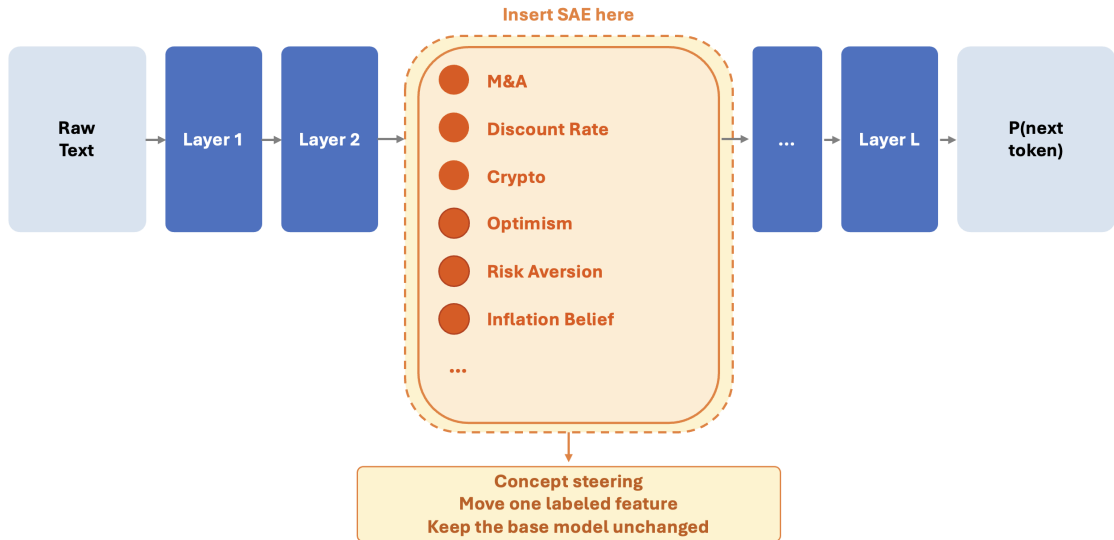
SAE as a dispersive prism



SAE as a dispersive prism



SAE as a dispersive prism



Economic theme map

Economic themes

- Macro-Finance & Economics
- Financial Markets & Instruments
- Corporate Finance
- Investment & Portfolio Management
- Risk Management & Derivatives

Examples of topics

- Inflation, GDP, monetary policy, yield curve
- Equity markets, liquidity, IPO, short selling
- M&A, governance, leverage, CAPEX
- Alpha, beta, Sharpe ratio, risk aversion
- Hedging, VaR, systemic risk, stress testing

This turns the 131k-dimensional sparse code into an economically named coordinate system.

Semantic validation: word-cloud probe

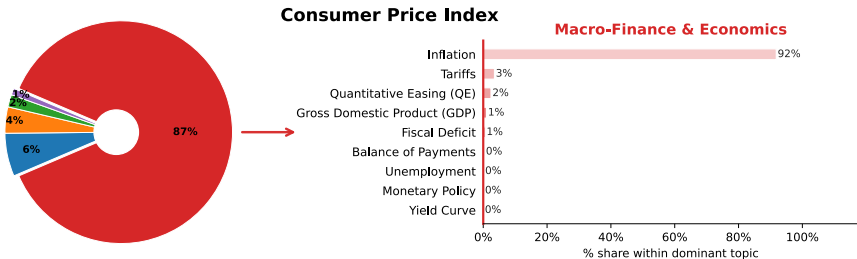
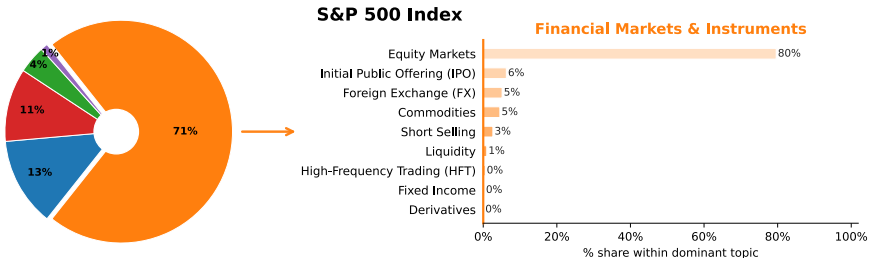
Probe

For each topic i , steer the top 10 features one at a time with coefficient $s = 200$, then ask:

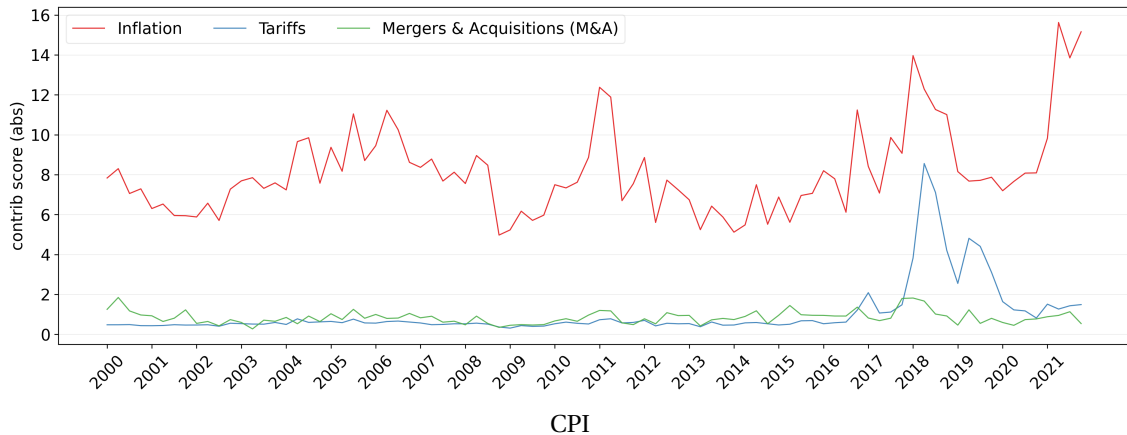
Give me 3 finance/economic concepts without explanation.

- Pool generated tokens.
- Remove stop words and render word frequencies.
- If labels are meaningful, topic-steered completions should use topic-related vocabulary.
- Word cloud represents the resulting frequency distribution.

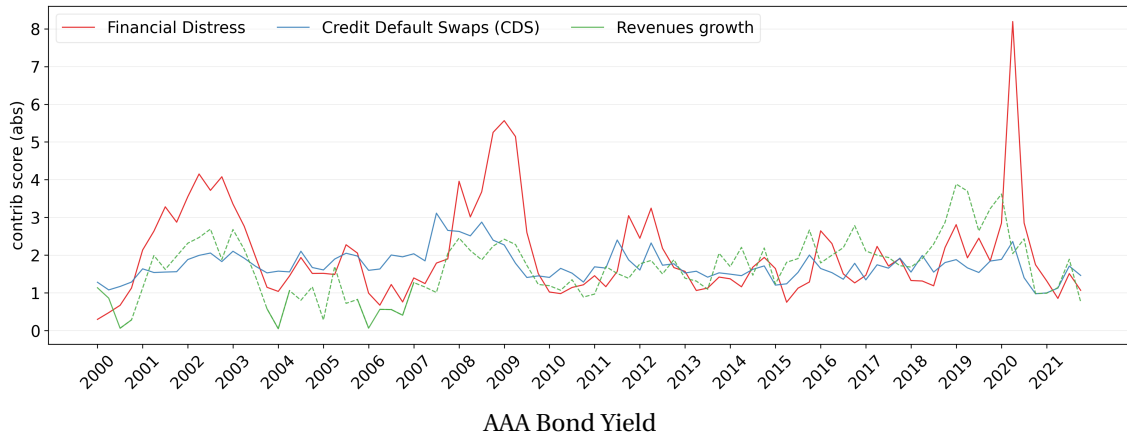
Which concepts drive macro expectations?



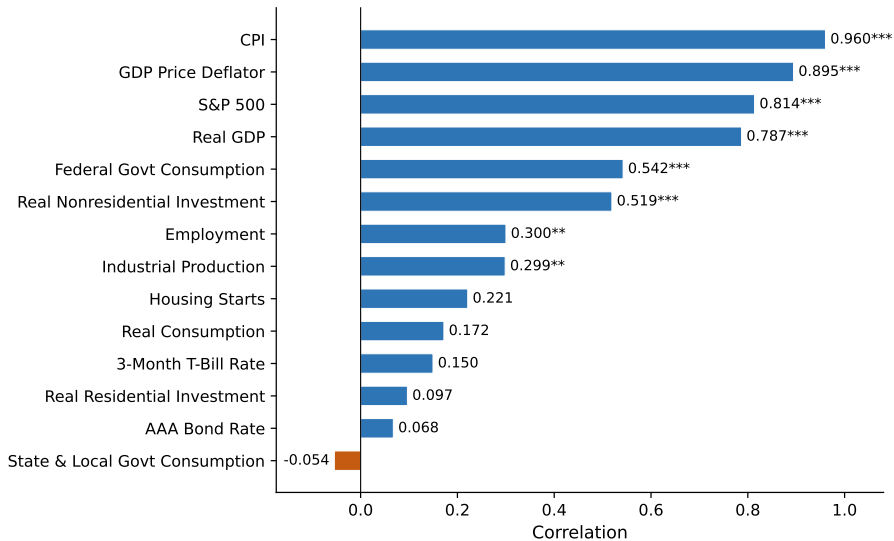
Which concepts drive macro expectations?



Which concepts drive macro expectations?



Mechanistic versus self-reported topic decomposition

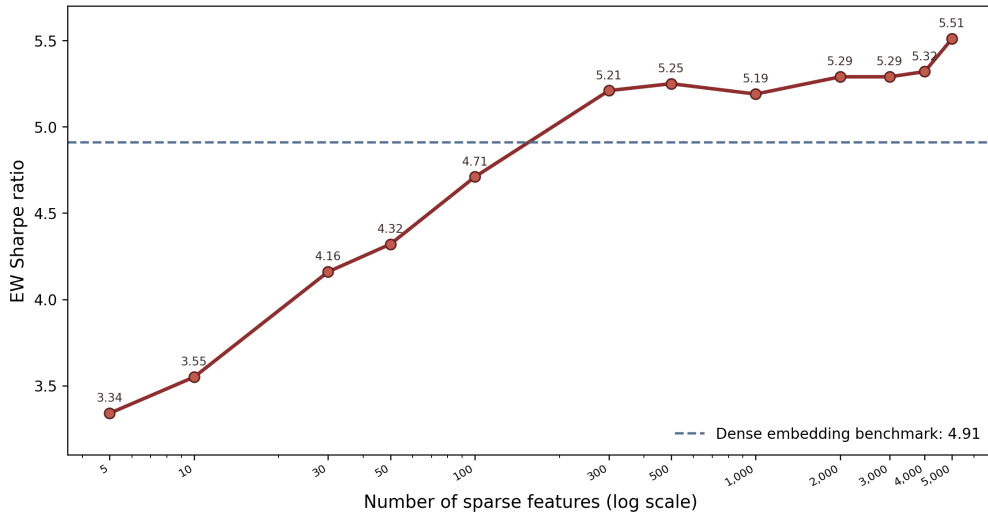


Interpretable embeddings and trading performance

Reuters firm news, 2015–2024:

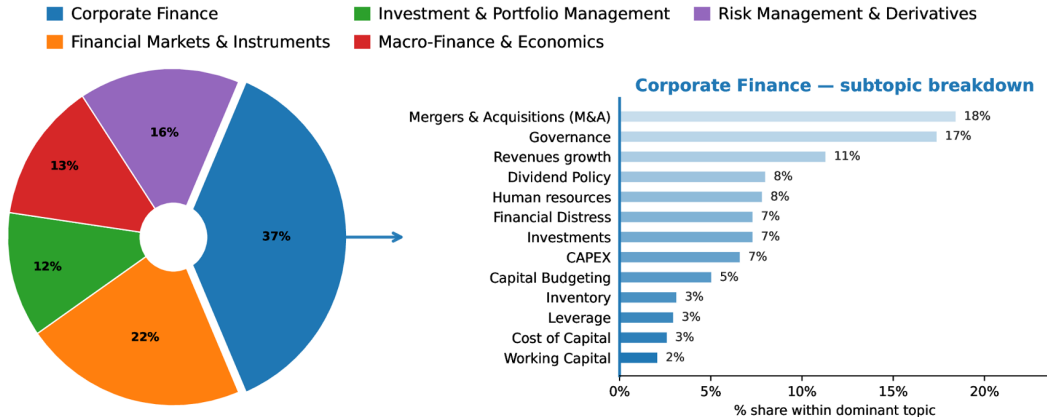
- 1 3.66 million articles about individual U.S.-listed firms, matched to CRSP returns.
- 2 Extract SAE vectors $v_{i,t}$ for each article.
- 3 Use a four-year rolling window to predict the sign of next intraday returns.
- 4 Build a long-short portfolio: long top 20% predicted positives, short bottom 20%.
- 5 Compare against dense last-layer embeddings from the same LLM.

Interpretability does not come at the cost of performance

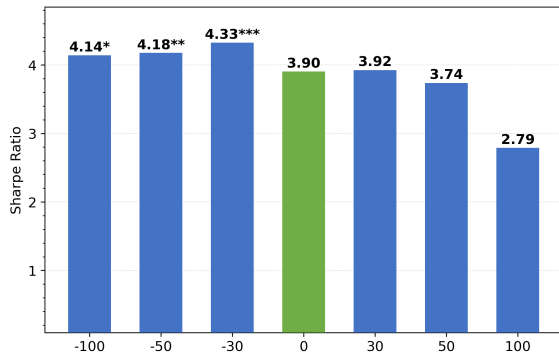


Alpha auditing

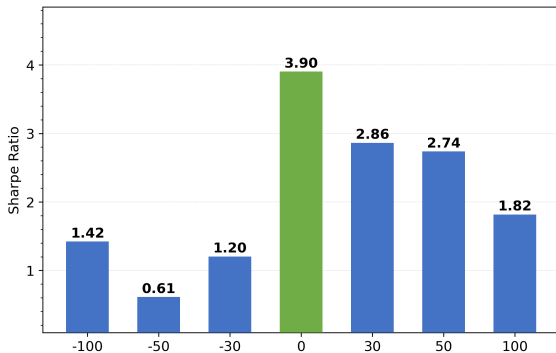
Which topics are responsible for the model's forecasting power?



SAE steering to remove optimism bias



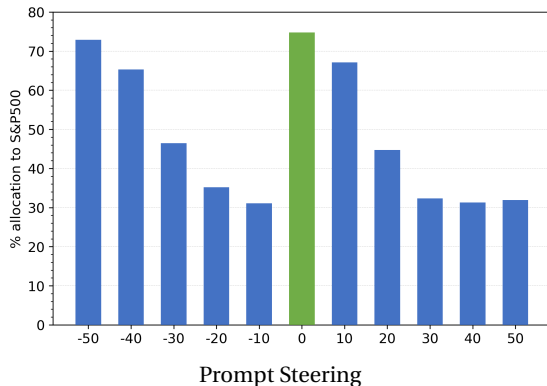
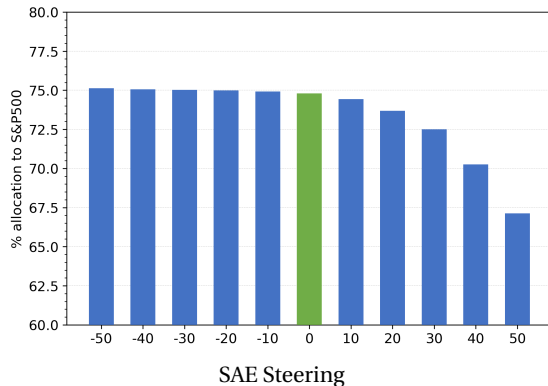
SAE steering



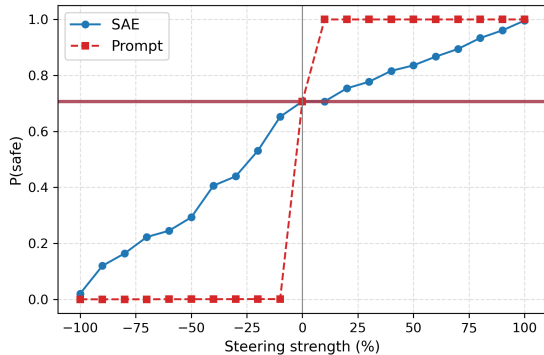
Prompt steering

- Main contrast: latent-feature steering to reduce sentiment enhances performance; prompt steering is unstable and performance collapses away from baseline.

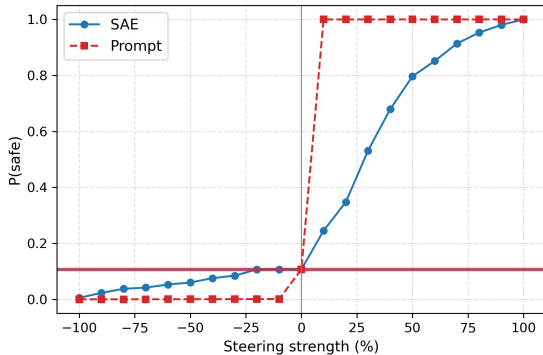
Steering the LLM's risk appetite for portfolio allocation



Steering the LLM's risk appetite for portfolio allocation



S&P500/Treasuries



Startup/Bond

SAE Use Cases

Interpretability

- Understand what economic concepts drive a decision
- Produce a better audit trail for reviewers
- Detect optimism, sentiment, framing bias

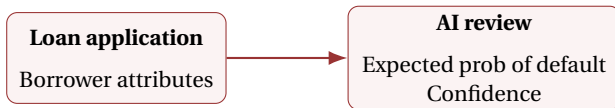
Steering

- De-bias pre-trained LLMs
- Scenario analysis with explicit reasoning shifts
- Silicon sampling

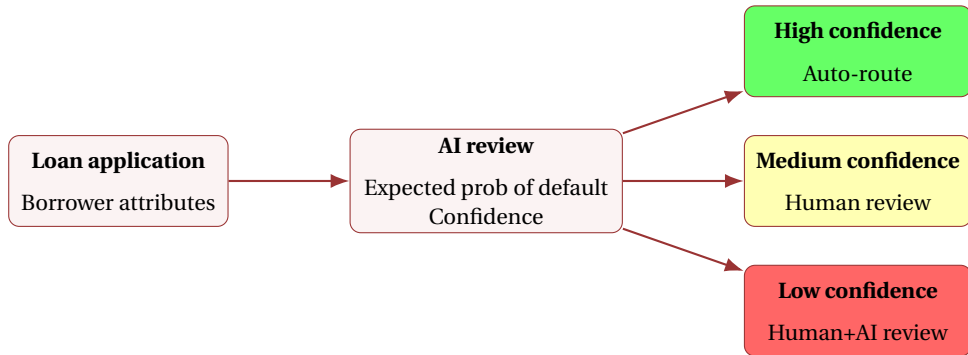
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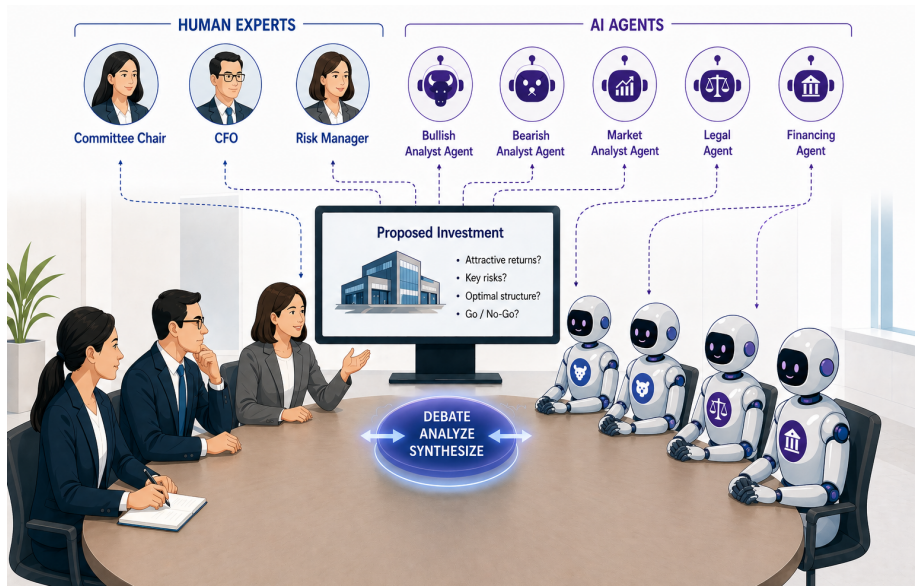
Human-AI collaboration in loan underwriting



Human-AI collaboration in loan underwriting



Your investment committee in the age of AI



Corporate investment committee example

Possible debate participants

- Bullish project analyst
- Bearish project analyst
- Market analyst
- Financing expert
- Risk manager
- Legal / compliance expert
- Human decision sponsor

What the controller can do

- Bring in risk manager earlier if the debate gets too optimistic
- Escalate legal expert when downside scenarios widen
- Delay expensive human review when consensus is already strong
- Create a clean record of what changed the decision

Outline

- 1 Why Not the Black Box?
- 2 When To Trust the LLM?
- 3 Opening the Black Box
- 4 Human In the Loop
- 5 Closing

Three takeaways

- 1 **Confidence:** Know when (not) to trust an AI system.
- 2 **Control:** AI systems can be inspected, diagnosed, and steered.
- 3 **Human-AI collaboration:** Take advantage of human expertise and AI strength.